

Research That Shapes First Volo Morphology

How research informs the learning design

Research-informed, standards-aligned supplemental morphology practice

Why morphology?

Morphology makes the meaningful structure of words visible. Research supports explicit attention to roots and bases, prefixes, suffixes, and the relationships among words in a family. Morphology instruction shows positive effects most consistently on morphological knowledge and word-level reading and spelling outcomes, while broader transfer - especially to reading comprehension - is less certain. (Bowers, Kirby, & Deacon, 2010; Colenbrander et al., 2024)

What the research points to

Make meaningful parts explicit

Students benefit from learning to identify and interpret morphemes rather than treating complex words as unanalyzed wholes. (Bowers et al., 2010; Mendes & Kirby, 2024)

Move from recognition to production

Instruction can progress from finding word parts to decomposition, word building, family reasoning, and selecting or producing appropriate derived forms. (Wolter & Green, 2013; Collins, 2023)

Treat morphology as multidimensional

Meaning, grammatical function, word form, spelling, and pronunciation are related but distinct aspects of morphological knowledge. (Goodwin, Petscher, & Tock, 2020; Apel, Henbest, & Petscher, 2023)

Use context as a partner

Students can use word structure to generate a meaning hypothesis and then use sentence or text context to confirm, reject, or refine it. (Wolter & Green, 2013; Mendes & Kirby, 2024)

First Volo's role. First Volo Morphology is designed as supplemental practice. It draws on research-supported instructional principles; the First Volo program itself has not been evaluated in a controlled efficacy trial.

Explicit word structure: find it, understand it, build with it

Across reviews and intervention studies, a recurring instructional pattern is to teach morphemes explicitly and then give students repeated opportunities to recognize, analyze, manipulate, and construct morphologically complex words. (Bowers et al., 2010; Mendes & Kirby, 2024; Wolter & Green, 2013)

Decomposition and construction belong together

Mendes and Kirby's Grade 4-6 intervention taught students what morphemes are, how free and bound bases work, how affixes contribute meaning, and how words can be assembled from or deconstructed into morphemes. Students also worked with Greek and Latin bases, word sums, word matrices, and morphological problem solving. (Mendes & Kirby, 2024)

Problem solving is more than memorizing a definition

Wolter and Green describe a "detective" approach in which students identify patterns, deconstruct complex words, reason by analogy, build words, and apply morphology while reading. Collins likewise describes analysis, synthesis, decomposition, morphological judgment, production/cloze tasks, and word-family work as practical intervention options. (Wolter & Green, 2013; Collins, 2023)

How this shapes First Volo

Learn + Meaning - teach a morpheme and connect it to useful words.

Find + Word Hunt - recognize a meaningful part across words and families.

Word Part + Break It Apart - identify the structure of a complex word rather than relying on visual resemblance alone.

Figure It Out - use known morphemes to generate a plausible whole-word meaning.

Build Words - combine meaningful parts and consider whether the resulting word is real and meaningful.

Digital practice can support this work

In a randomized six-week pilot with 263 third graders, computer-delivered morphology-focused instruction produced significant gains in affix identification and suffix choice. The program included explicit derivational morphology, roots and bases, word construction, decomposition, and word matrices. Not every measured morphology or spelling outcome improved, so the findings support targeted digital morphology practice rather than broad claims about literacy outcomes. (Wood et al., 2025)

Derivational morphology: meaning, grammar, and the right word form

Knowing a root or suffix meaning is only part of morphological knowledge. Students also need to understand how derivational morphemes change a word's grammatical role and how related forms function differently in sentences. (Goodwin et al., 2020; Wolter & Green, 2013)

Morphology has multiple dimensions

In 1,140 students in Grades 5-8, Goodwin, Petscher, and Tock examined morphological awareness along with syntactic, semantic, and phonological/orthographic morphological knowledge. The dimensions related differently to reading comprehension, supporting instruction and progress monitoring that do not collapse morphology into one undifferentiated skill. (Goodwin et al., 2020)

Task and word form matter

Across large Grade 3-6 samples, base-word transparency and task type were important predictors of performance. Words with spelling and/or pronunciation changes were harder than transparent relationships. (Apel et al., 2023)

Teach into sentence-level choices

Wolter and Green explicitly connect derivational suffixes to parts of speech and sentence slots. Henbest, Apel, and Mitchell used tasks in which a base word had to be modified so the resulting word fit an incomplete sentence. Wood et al. also found a significant treatment effect on a suffix-choice task that required selecting the morphologically appropriate form. (Wolter & Green, 2013; Henbest, Apel, & Mitchell, 2019; Wood et al., 2025)

EXAMPLE OF THE PLANNED CHANGE IT PROGRESSION

Teach: *-tion / -sion* often forms a noun naming an action, process, or result.

Family: predict → prediction → predictive → predicted

Apply: The scientist made an accurate _____.
predict predicted **prediction** predictive

The goal is not guessing from context alone. Students first learn what the word parts and forms do, then use morphology, grammar, and sentence meaning together.

learn the function → compare family members → choose the form → eventually produce the form

From words to sentences - and then to connected text

Research-informed morphology practice does not have to stop at isolated words. Several instructional sources describe applying morphological analysis while reading, using context to check a proposed meaning, and locating related word structures in connected text.

(Wolter & Green, 2013; Mendes & Kirby, 2024)

Morphology first, context as a check

Wolter and Green describe students using morphological information to infer an unfamiliar word's meaning and then using surrounding text to decide whether that interpretation makes sense. Mendes and Kirby included texts with target morphemes, asked students to identify and deconstruct multimorphemic words, and encouraged them to look for words with similar structures.

(Wolter & Green, 2013; Mendes & Kirby, 2024)

How this shapes sentence and text activities

Use It - choose the word whose morphology and whole-word meaning fit a sentence.

Change It - choose or produce the member of a morphological family whose grammatical role fits the sentence.

Text Detective - read a short passage, locate a morphologically complex word, break it apart, infer a meaning from the parts, and check the interpretation against the text.

WHAT TEXT DETECTIVE CAN LOOK LIKE

Passage: The scientists **reconstructed** the ancient building from drawings. They studied each section carefully before creating a model.

Analyze: re + struct + ed

Infer: *re-* = again; *struct* = build → **built again**

Check the text: Does that meaning fit what the scientists did?

A useful boundary

Connected-text morphology practice is an application of word analysis in context. The current intervention literature does not justify treating morphology instruction by itself as a reliable route to broad reading-comprehension gains. The 2024 systematic review found positive immediate effects on reading and spelling overall, but no clear immediate effect on reading comprehension. (Colenbrander et al., 2024)

Transparency, transfer, and thoughtful progression

The difficulty of a morphology task depends on more than a student's grade or whether a word contains an affix. Research supports paying attention to the transparency of the relationship, the task itself, and whether the student is working with a taught word or transferring knowledge to a new word.

(Apel et al., 2023; Colenbrander et al., 2024)

Transparency

Apel and colleagues distinguished transparent relationships from words involving phonological shifts, orthographic shifts, or both. Transparency and task type contributed substantially to item difficulty. (Apel et al., 2023)

Transfer

The 2024 systematic review found the clearest evidence for words directly taught during instruction. Evidence for transfer to untrained words was more limited, especially for reading, although some spelling transfer was found. (Colenbrander et al., 2024)

Near transfer is still important

Mendes and Kirby deliberately assessed both taught and untaught morphologically complex words. Their small Grade 4-6 study found gains on morphology-specific measures that were not confined to the words practiced during intervention, even though standardized word-reading, fluency, and spelling measures did not show significant treatment effects. (Mendes & Kirby, 2024)

How this shapes First Volo's progression

- **Practice band, vocabulary accessibility, and morphological transparency are treated as separate decisions.**
- More transparent examples can introduce a pattern before students encounter spelling or pronunciation shifts.
- Progress monitoring can distinguish performance on **taught words** from performance on **transfer words** containing familiar morphemes.
- Different activities sample different demands: recognition, meaning, decomposition, word building, grammatical form, and contextual application.

Designed for use by educators and SLPs

Henbest, Apel, and Mitchell found that two second-grade teachers were able to implement a scripted, scaffolded morphology program with brief SLP support and high implementation fidelity; students improved on all four morphology measures, although the feasibility study had no business-as-usual comparison group. (Henbest et al., 2019) Wood et al. similarly demonstrated that supplemental computer-delivered morphology practice could be implemented in authentic third-grade classrooms. (Wood et al., 2025)

Standards and literacy guidance

Common Core

Common Core language standards ask students to use context and meaningful word parts to determine word meaning. Expectations for prefixes, suffixes, roots, and Greek and Latin word elements continue across the elementary and middle grades.

State literacy guidance

Massachusetts DESE places vocabulary and morphology within language comprehension and recommends explicit attention to word meanings, word parts, spelling, roots, prefixes, suffixes, and meaningful context. California foundational-skills guidance likewise connects morphemes with decoding and meaning.

Key references

- Apel, K., Henbest, V. S., & Petscher, Y. (2023). Effects of affix type and base word transparency on students' performance on different morphological awareness measures. *Journal of Speech, Language, and Hearing Research*, 66, 239-256.
- Bowers, P. N., Kirby, J. R., & Deacon, S. H. (2010). The effects of morphological instruction on literacy skills: A systematic review of the literature. *Review of Educational Research*, 80(2), 144-179.
- Colenbrander, D., von Hagen, A., Kohnen, S., Wegener, S., Ko, K., Beyersmann, E., Behzadnia, A., Parrila, R., & Castles, A. (2024). The effects of morphological instruction on literacy outcomes for children in English-speaking countries: A systematic review and meta-analysis. *Educational Psychology Review*, 36, 119.
- Collins, G. G. (2023). Morphological interventions to support literacy from kindergarten to grade 12. *Perspectives of the ASHA Special Interest Groups*, 8(6), 1205-1219.
- Goodwin, A. P., Petscher, Y., & Tock, J. (2020). Morphological supports: Investigating differences in how morphological knowledge supports reading comprehension for middle school students with limited reading vocabulary. *Language, Speech, and Hearing Services in Schools*, 51(3), 589-602.
- Green, C., & Coxhead, A. (2026). The Vocabulary Teachers' Competencies (VTC) framework: The core knowledge, skills and professional capabilities that support effective vocabulary instruction. *Literacy*, 60(1), e70016.
- Henbest, V. S., Apel, K., & Mitchell, A. (2019). Speech-language pathologist-guided morphological awareness instruction in the general education classroom. *Perspectives of the ASHA Special Interest Groups*, 4(5), 771-780.
- Mendes, B. B., & Kirby, J. R. (2024). The effects of a morphological awareness intervention on reading and spelling ability of children with dyslexia. *Learning Disability Quarterly*, 47(4), 222-233.
- Wolter, J. A., & Green, L. (2013). Morphological awareness intervention in school-age children with language and literacy deficits: A case study. *Topics in Language Disorders*, 33(1), 27-41.
- Wood, C., Torres-Chavarro, M., Tibi, S., Schatschneider, C., & Ke, F. (2025). Computer-delivered morphology-focused vocabulary instruction: A pilot study. *American Journal of Speech-Language Pathology*, 34, 818-833.
- Massachusetts Department of Elementary and Secondary Education. (2022). *Mass Literacy: Vocabulary and Morphology (Language Comprehension)*.
- California Department of Education. (2015). *Resource Guide to the Foundational Skills of the California Common Core State Standards for English Language Arts and Literacy*.

About this summary. These are key sources that directly inform the design of First Volo Morphology. The research base is broader than the references listed here. Updated August 2026.

Trusted guidance & further reading

These public resources provide additional context for educators and SLPs who want to explore how morphology fits within reading, vocabulary, and language instruction. They complement the peer-reviewed studies summarized in this document.

U.S. Department of Education - What Works Clearinghouse

Providing Reading Interventions for Students in Grades 4-9

The practice guide includes explicit work with prefixes and suffixes, base-word and affix analysis, Greek and Latin roots, related word forms, and practice with complex words in sentences and passages.

[Open WWC practice guide](#)ies.ed.gov/ncee/WWC/PracticeGuide/29

Massachusetts DESE - Mass Literacy

Vocabulary and Morphology

Massachusetts guidance places vocabulary and morphology within language comprehension and highlights explicit attention to word meanings, word parts, spelling, roots, prefixes, suffixes, and meaningful context.

[Open Mass Literacy resource](#)www.doe.mass.edu/massliteracy/skilled-reading/language-comprehend/vocab-mor...

AdLit

Key Literacy Component: Morphology

An educator-friendly overview of morphemes, derivational and inflectional morphology, Greek and Latin word parts, and practical connections to adolescent reading and vocabulary instruction.

[Open AdLit morphology guide](#)www.adlit.org/topics/phonics-word-study-decoding/key-literacy-component-mor...

How to use these links

These resources are included as trusted guidance and further reading. They help place morphology within broader literacy practice; they are not evaluations or endorsements of First Volo Morphology.